

Supplemental Material for “The UZH protocol: Separating and reducing Gaussian-basis and pseudopotential errors in CP2K”

Hossein Mirhosseini,^{1,2} Tiziano M. A. Müller,³ Matthias
Krack,⁴ Thomas D. Kühne,^{1,2,5,*} and Jürg Hutter³

¹*Center for Advanced Systems Understanding (CASUS),
Conrad-Schiedt-Straße 20, 02826 Görlitz, Germany*

²*Helmholtz-Zentrum Dresden-Rossendorf,
Bautzner Landstraße 400, 01328 Dresden, Germany*

³*Department of Chemistry, University of Zurich, CH-8057 Zurich, Switzerland*

⁴*Paul Scherrer Institute, 5232 Villigen PSI, Switzerland*

⁵*Institute of Artificial Intelligence, Technische Universität Dresden,
Helmholtzstraße 10, 01069 Dresden, Germany*

I. SOURCE MATERIAL AND IMPLEMENTATION DETAILS

The main text summarizes the protocol, while this Supplemental Material records additional details that are useful for reproduction and for future development of the UZH settings. The organization follows the original molecularly optimized (MOLOPT), Goedecker–Teter–Hutter (GTH), and Hartwigsen–Goedecker–Hutter (HGH) literature [1–3], the revised MOLOPT benchmarking work [4], the CP2K implementation papers [5, 6], and the Automated Interactive Infrastructure and Database for Computational Science (AiiDA) verification protocol and its Supplemental Information [7, 8].

A. CP2K ATOM and basis optimization

The updated CP2K ATOM implementation developed for the present UZH work is exposed through a dedicated input section for atomic calculations. The relevant run types are energy calculations, basis optimization, and pseudopotential optimization. The ATOM method section supports restricted and unrestricted Kohn-Sham and Hartree-Fock atomic calculations and scalar-relativistic Hamiltonians including zeroth-order regular approximation (ZORA)-type and Douglas–Kroll–Hess variants. The relativistic section allows the order of the Douglas–Kroll–Hess transformation, the transformation strategy, and the ZORA model-potential variant to be selected. The potential section supports all-electron, GTH, unified pseudopotential format, separable Gaussian pseudopotential, and effective core potential inputs. For the GTH format, the input layer exposes a CP2K-standard GTH potential block, and the print layer can write the optimized potential in CP2K format and in a form compatible with the unified pseudopotential format.

The CP2K basis optimizer works from template, work, and output basis files. It supports simultaneous fitting of related basis levels and allows the condition number of the overlap matrix to enter the objective. This mirrors the original MOLOPT design: diffuse flexibility is desirable for weak interactions and response properties, but the final basis must remain numerically stable in condensed-phase simulations.

The pseudopotential optimizer constructs both all-electron and pseudopotential atomic references. Occupations are partitioned into core and valence contributions; semicore choices

* tkuehne@cp2k.org

are therefore explicit in the input and can be changed when the verification workflow reveals poor transferability. The optimizer evaluates the all-electron reference with the selected scalar-relativistic Hamiltonian, constructs the corresponding pseudopotential atom, and fits the compact GTH parameters with a derivative-free Powell algorithm. The fitted target is therefore not an isolated numerical object but a scalar-relativistic atomic mapping from an all-electron reference to the analytic separable GTH form. This machinery is what allows the UZH protocol to improve the pseudopotential component once the SIRIUS-GTH-UZH versus SIRIUS-FP-LAPW comparison identifies a pseudopotential-limited element.

B. SIRIUS calculations

SIRIUS is used in two complementary modes [9–11]. In pseudopotential plane-wave mode, SIRIUS-GTH-UZH uses the same GTH-UZH pseudopotentials as CP2K but removes the localized Gaussian-basis approximation. In all-electron full-potential linearized augmented-plane-wave (FP-LAPW) mode, SIRIUS provides the reference equation of state. The CP2K-GTH-UZH versus SIRIUS-GTH-UZH comparison isolates the CP2K Gaussian-basis error, while the two SIRIUS modes isolate the pseudopotential error.

In the FP-LAPW calculation the unit cell is divided into non-overlapping muffin-tin spheres and an interstitial region. Radial functions multiplied by spherical harmonics describe the wave functions inside the spheres, while plane waves describe the interstitial region; local orbitals are used to improve the description of semicore and high-lying states. Because the method is full-potential, the charge density and Kohn-Sham potential are represented without imposing a muffin-tin shape approximation. This all-electron reference is therefore well suited for the present purpose: any difference between SIRIUS-GTH-UZH and SIRIUS-FP-LAPW primarily measures the frozen-core and pseudization error of the GTH-UZH potential, including the quality of semicore and scalar-relativistic transferability.

For heavy elements the relativistic settings are part of the definition of the reference. The SIRIUS full-potential species files contain the radial grid, linearized augmented-plane-wave (LAPW)/augmented-plane-wave plus local-orbital (APW+lo) basis information, core-state treatment, and valence-relativity choice. The FP-LAPW reference calculations use the infinite-order regular approximation (IORA) for valence scalar relativity [12]; this IORA setting is recorded in the full-potential species definition and kept fixed throughout the

equation-of-state comparison. The CP2K interface exposes the electronic-structure mode as either pseudopotential or full-potential LAPW+lo, and the full-potential setup distinguishes core relativity from valence relativity. In the UZH protocol we compare scalar-relativistic pseudopotentials to scalar-relativistic all-electron references; explicit spin-orbit splittings are outside the present benchmark. This convention keeps the error decomposition aligned with the scalar-relativistic GTH/HGH pseudopotentials used by CP2K/Quickstep.

II. MOLECULAR CALIBRATION

The molecular calibration assesses the revised UZH-MOLOPT basis sequence before the solid-state verification step. The benchmark contains small molecules covering main-group bonds, polar bonds, non-linear and dihedral angles, dipole moments, polarizabilities, and vibrational frequencies. The underlying CP2K molecular test calculations are provided in the MolTest Zenodo archive, which contains calculations with revised CP2K basis sets and pseudopotentials and CP2K all-electron basis sets for Perdew–Burke–Ernzerhof (PBE), Perdew–Burke–Ernzerhof hybrid (PBE0), and Tao–Perdew–Staroverov–Scuseria (TPSS) exchange–correlation functionals [13–16]. Throughout the molecular statistics, DZVP, TZVP, and TZV2P refer to CP2K/GPW calculations with the corresponding GTH-UZH pseudopotentials. The SVP, TZVPP, and QZVPP entries denote CP2K/GAPW all-electron calculations with corresponding UZH-MOLOPT basis sets, which are also part of the UZH protocol; they follow the def2-type contraction hierarchy but are not Gaussian 16 calculations. Gaussian 16/def2-QZVP provides the external molecular reference used to define the plotted relative errors. Figures 1, 2, and 3 provide the angular and dipole-moment statistics that complement the main-text bond-length and polarizability analysis. Figures 4 and 5 summarize two additional molecular checks from the thesis-level calibration data: the bond-class dependence of the structural errors and the consistency of Gaussian 16 and CP2K all-electron reference calculations. The role of this stage is to ensure chemically balanced transferability and numerical stability before the basis is used in periodic calculations.

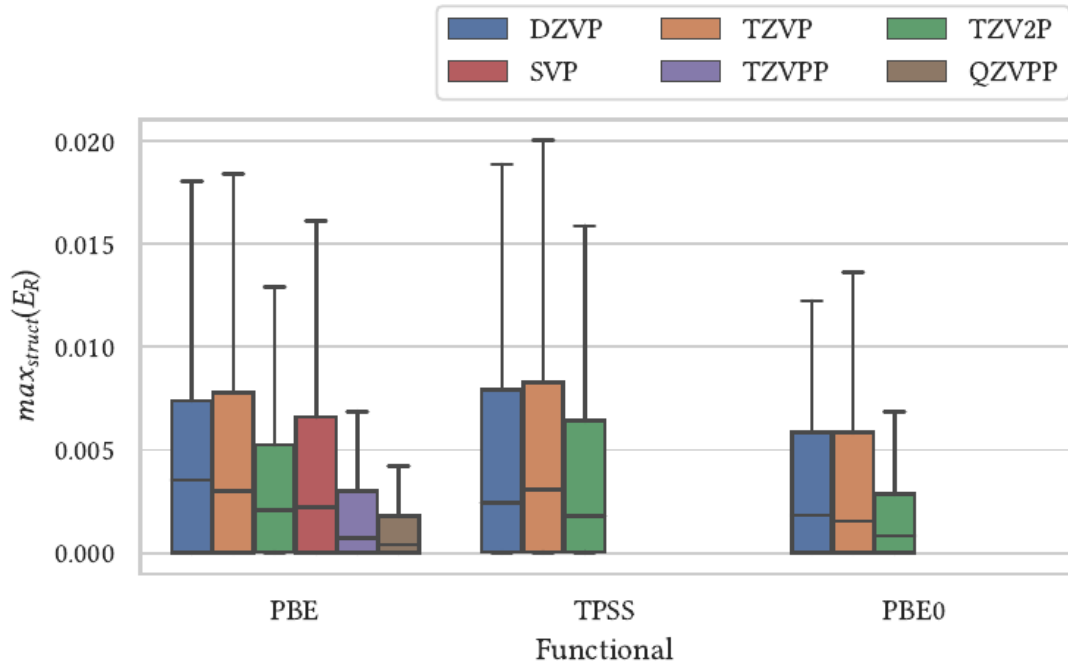


FIG. 1. Distribution of relative errors for nonlinear angles in the molecular calibration set. Angular observables are more sensitive than bond lengths to automated atom matching and near-symmetric geometries, but the systematic improvement from DZVP to TZV2P remains visible.

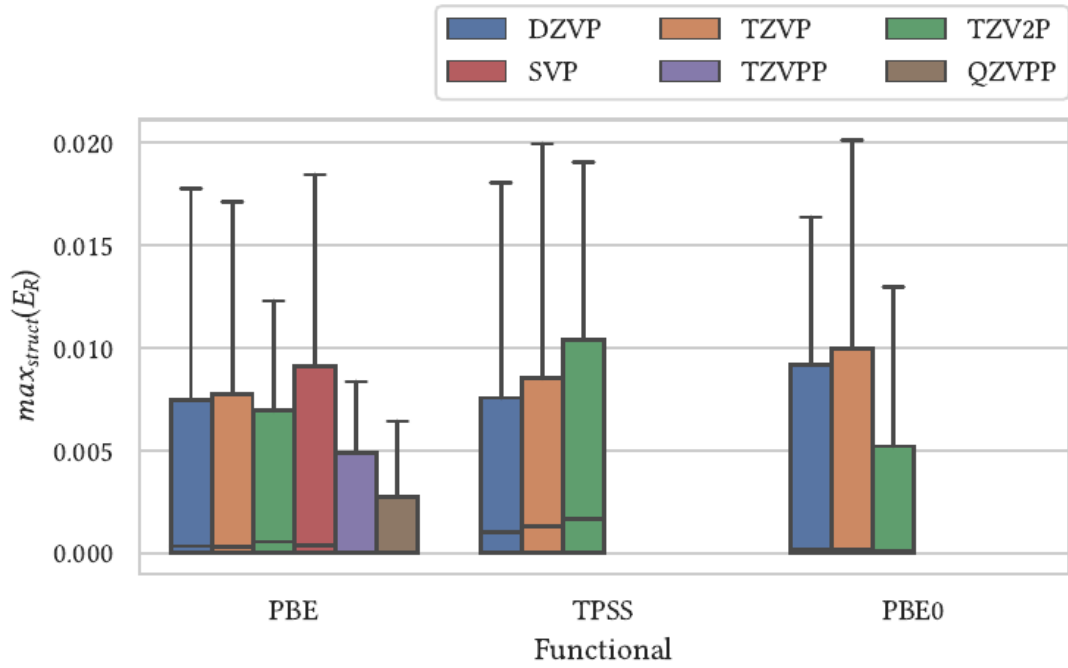


FIG. 2. Distribution of relative errors for dihedral angles in the molecular calibration set. The broader tails reflect the stronger sensitivity of torsional coordinates to small structural and matching differences, while the basis-set hierarchy remains consistent with the bond-length and nonlinear-angle tests.

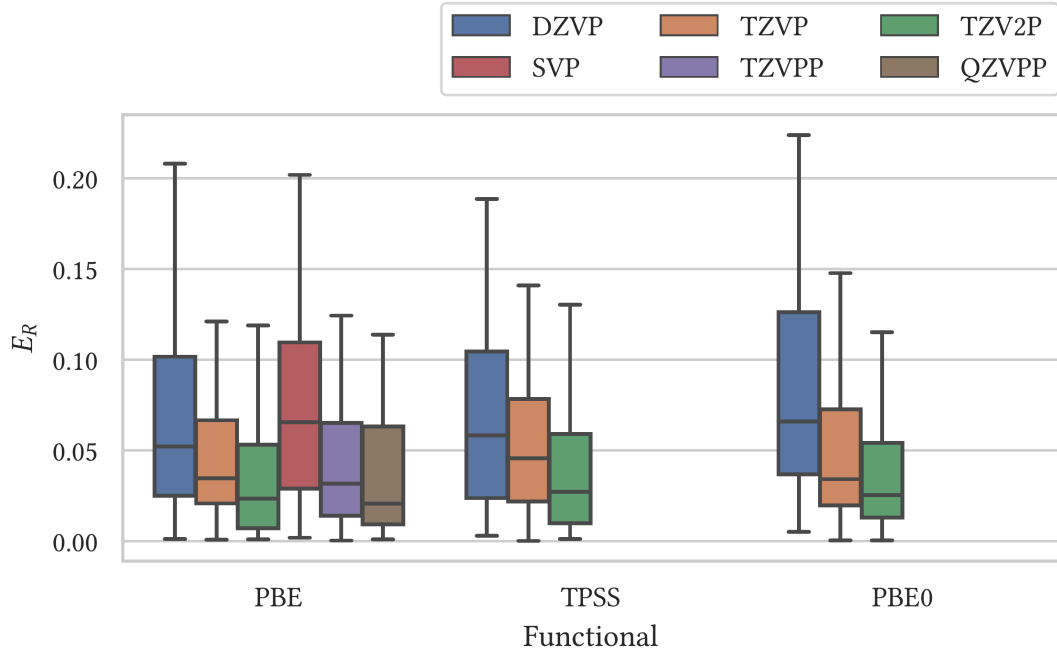


FIG. 3. Distribution of relative errors for molecular dipole moments, adapted from Fig. 4.11 of Ref. [4]. The statistics include converged structures with non-vanishing dipole moments and show that response-sensitive density properties retain the same systematic basis-set improvement observed for structural observables.

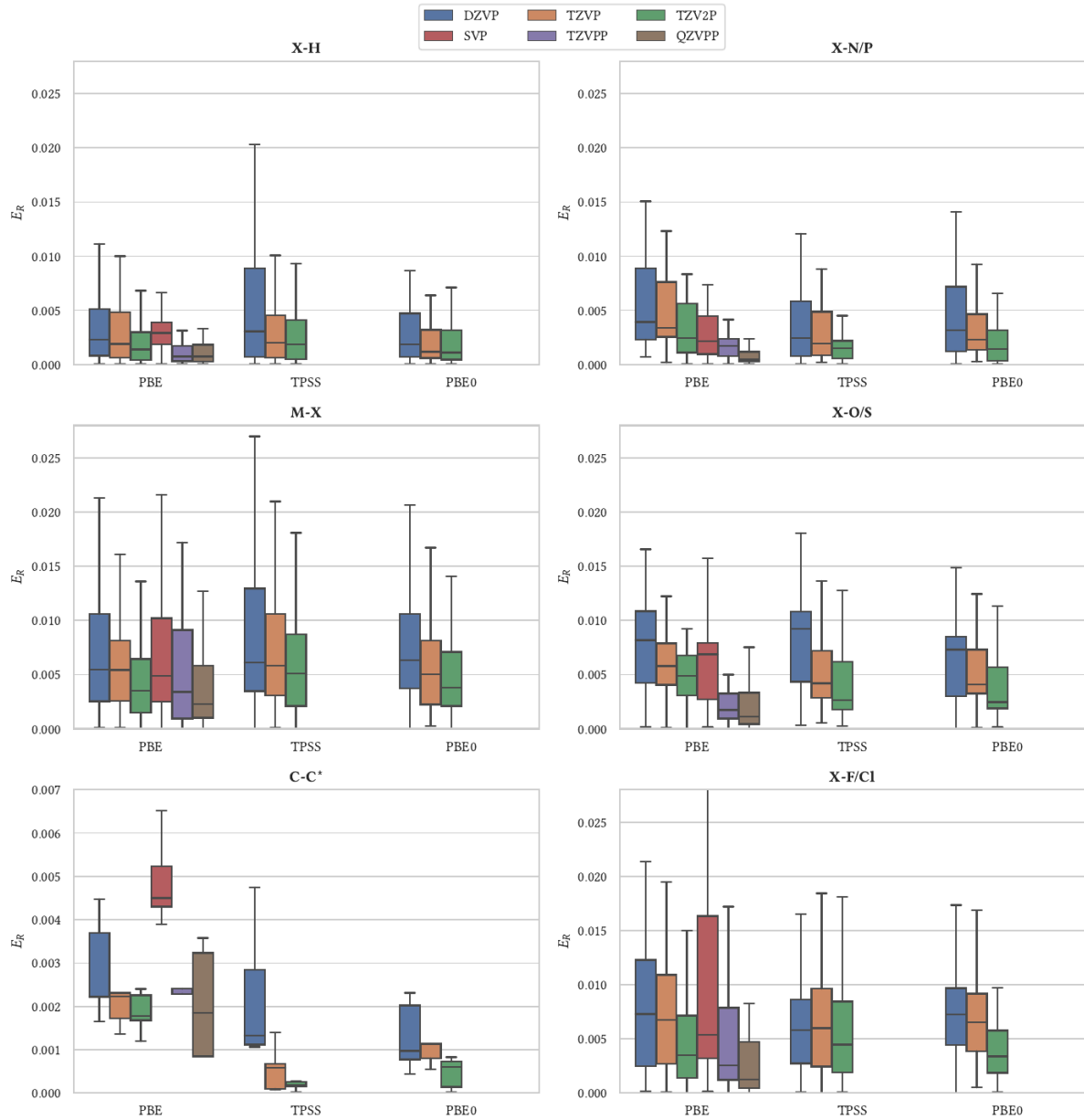


FIG. 4. Bond-class-resolved distribution of relative bond-length errors across the molecular calibration set, adapted from Fig. 4.13 of Ref. [4]. Outliers are hidden and the C–C panel uses a smaller vertical scale. The grouping shows that the UZH-MOLOPT basis sequence improves systematically across chemically distinct bond classes before periodic verification.

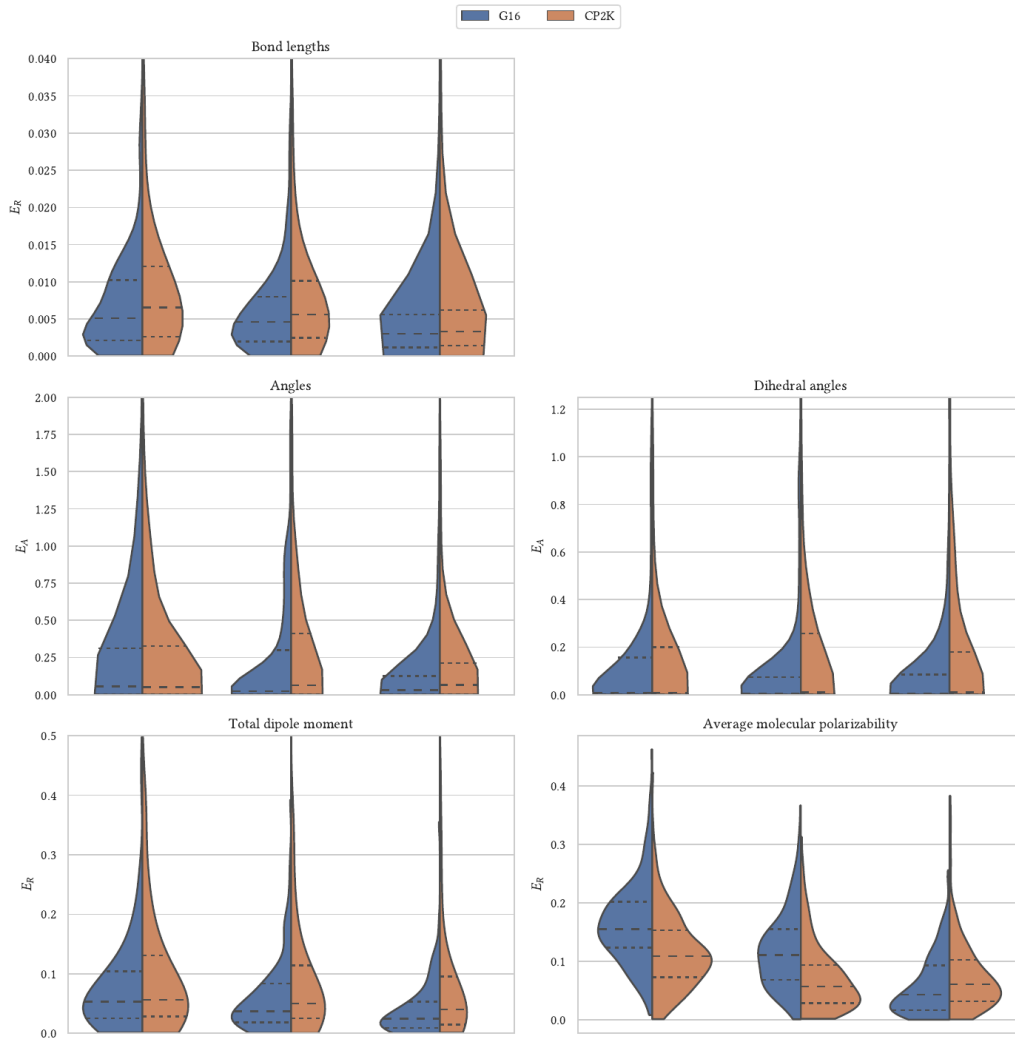


FIG. 5. Comparison of molecular calibration errors obtained with Gaussian 16/def2-QZVP and CP2K/GAPW-QZVPP all-electron references, adapted from Fig. 4.16 of Ref. [4]. Here QZVPP denotes the corresponding UZH-MOLOPT basis set used in the CP2K/GAPW all-electron calculation, while def2-QZVP denotes the external Gaussian 16 reference. Relative errors are shown for bond lengths, dipole moments, and average molecular polarizabilities, while absolute errors are shown for angles and dihedral angles. The close agreement supports the use of CP2K/GAPW as the internal all-electron branch of the UZH calibration loop.

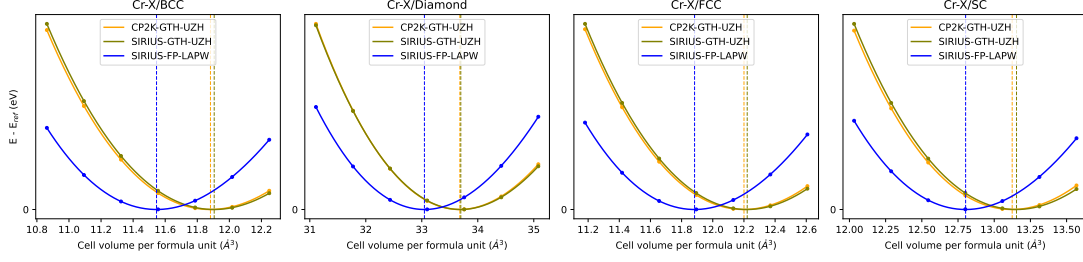


FIG. 6. Representative equation-of-state curves for Cr. The behavior is characteristic of a pseudopotential-limited case in the original settings.

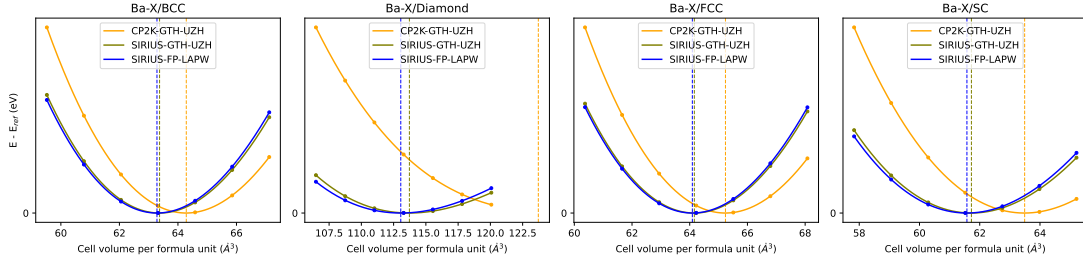


FIG. 7. Representative equation-of-state curves for Ba. The comparison highlights a case with a substantial Gaussian-basis contribution in the original CP2K-GTH-UZH settings.

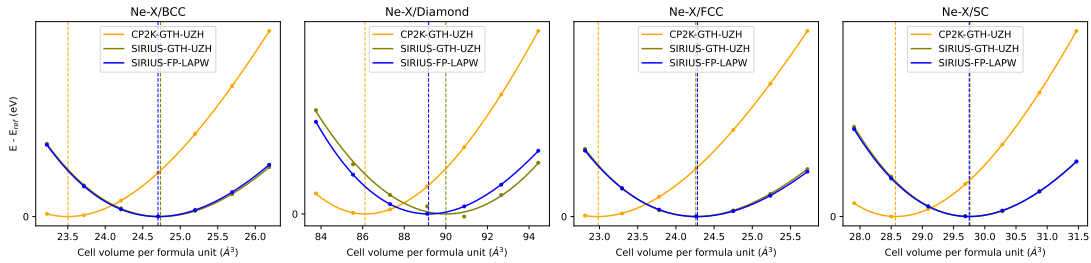


FIG. 8. Representative equation-of-state curves for Ne. Noble gases are sensitive tests of diffuse basis flexibility because their equation-of-state curves are shallow.

III. REPRESENTATIVE EQUATION-OF-STATE CURVES

The periodic-table heat maps in the main text are compact summaries of many equation-of-state calculations. Figures 6, 7, 8, 9, and 10 show representative raw equation-of-state comparisons for elements that illustrate different error regimes.

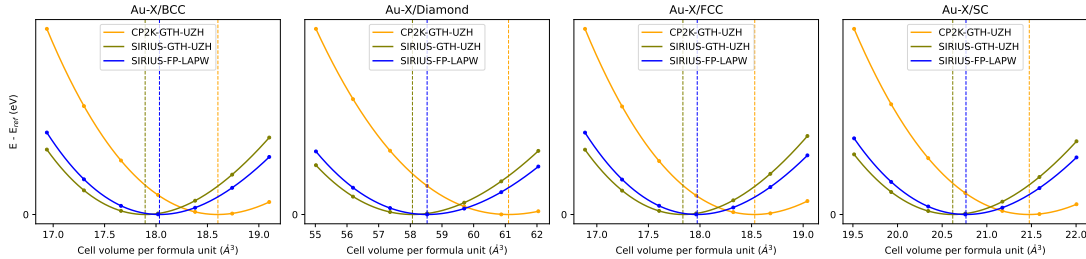


FIG. 9. Representative equation-of-state curves for Au, illustrating the behavior of a heavy element where basis completeness and relativistic pseudopotential transferability must both be monitored.

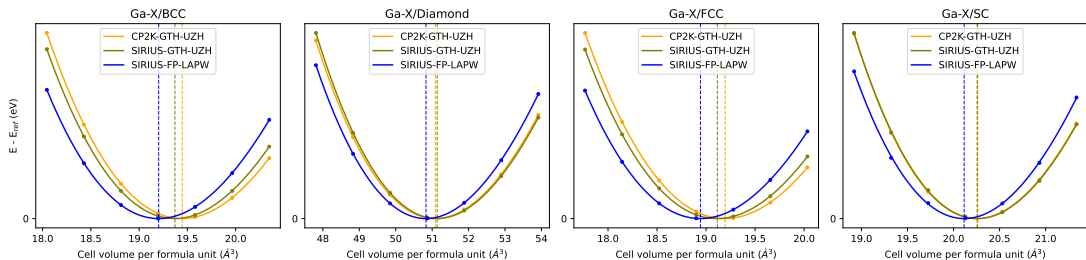


FIG. 10. Representative equation-of-state curves for Ga.

IV. RELEASE REPOSITORY ORGANIZATION

The release repository for the UZH protocol is hosted in the DCM-Uni-Paderborn GitHub organization under the repository name UZH-protocol, <https://github.com/DCM-Uni-Paderborn/UZH-protocol>. The deposited files have been organized according to the three comparisons used in the main text rather than copied as a flat working directory. The main repository folders are:

1. `cp2k-data/`, which contains the official CP2K UZH data files `BASIS_MOLOPT_UZH`, `BASIS_ADMM_UZH`, `BASIS_TB_UZH`, `BASIS_ccGRB_UZH`, and `POTENTIAL_UZH` taken from the CP2K source tree.
2. `data/cp2k-gth-uzh/`, which contains the CP2K-GTH-UZH equation-of-state result files, diagnostic plots, improved-basis reruns, focused resubmissions, and noble-gas augmented-basis tests.
3. `data/sirius-gth-uzh/`, which contains the SIRIUS-GTH-UZH original-setting results, improved-pseudopotential results, focused old/new pseudopotential compar-

isons, and an explicit Sr diamond input/output test case.

4. `data/sirius-fp-lapw/`, which contains the all-electron SIRIUS-FP-LAPW reference equation-of-state result file.
5. `figures/` and `manuscript/`, which contain the publication figures and the source files for the manuscript and Supplemental Material.
6. `manifests/`, which contains a source-path mapping, a file inventory, workflow-organization metadata, software/provenance metadata, and SHA-256 checksums for the deposited files.

The JSON result files contain the parsed equation-of-state data, Birch–Murnaghan fit data, UUID mappings, and workflow failure or missing-output diagnostics. The per-element PDF files are retained as equation-of-state diagnostics that connect the periodic-table maps in the main text to the underlying energy-volume curves. The repository therefore supports two complementary checks: the tabulated and plotted ε values can be traced to machine-readable result files, and individual outliers can be inspected directly at the equation-of-state level.

The present release already contains the periodic verification data required for the CP2K-GTH-UZH, SIRIUS-GTH-UZH, and SIRIUS-FP-LAPW comparisons discussed in the main text. The molecular calibration data are linked to the MolTest Zenodo archive cited in the main text and in Sec. II of this Supplemental Material. Additional raw molecular input/output tables, executable post-processing scripts for regenerating the periodic-table maps from the JSON result files, and final software-version metadata should be added before the repository is frozen for archival publication.

-
- [1] J. VandeVondele and J. Hutter, Gaussian basis sets for accurate calculations on molecular systems in gas and condensed phases, *The Journal of Chemical Physics* **127**, 114105 (2007).
 - [2] S. Goedecker, M. Teter, and J. Hutter, Separable dual-space gaussian pseudopotentials, *Physical Review B* **54**, 1703 (1996).
 - [3] C. Hartwigsen, S. Goedecker, and J. Hutter, Relativistic separable dual-space gaussian pseudopotentials from h to rn, *Physical Review B* **58**, 3641 (1998).

- [4] T. M. A. Müller, *From Benchmarking to Periodic Fock Exchange in the Auxiliary Density Matrix Method via k -Point Sampling with Gaussian Basis Sets*, Ph.D. thesis, University of Zurich (2024).
- [5] T. D. Kühne, M. Iannuzzi, M. Del Ben, V. V. Rybkin, P. Seewald, F. Stein, T. Laino, R. Z. Khaliullin, O. Schütt, F. Schiffmann, D. Golze, J. Wilhelm, S. Chulkov, M. H. Bani-Hashemian, V. Weber, U. Borstnik, M. TAILLEFUMIER, A. S. Jakobovits, A. Lazzaro, H. Pabst, T. Müller, R. Schade, M. Guidon, S. Andermatt, N. Holmberg, G. K. Schenter, A. Hehn, A. Bussy, F. Belleflamme, G. Tabacchi, A. Glöck, M. Lass, I. Bethune, C. J. Mundy, C. Plessl, M. Watkins, J. VandeVondele, M. Krack, and J. Hutter, Cp2k: An electronic structure and molecular dynamics software package – quickstep: Efficient and accurate electronic structure calculations, *The Journal of Chemical Physics* **152**, 194103 (2020).
- [6] M. Iannuzzi, J. Wilhelm, F. Stein, A. Bussy, H. Elgabarty, D. Golze, A.-S. Hehn, M. Graml, S. Marek, B. S. Gökmen, C. Schran, H. Forbert, R. Z. Khaliullin, A. Kozhevnikov, M. TAILLEFUMIER, R. Meli, V. V. Rybkin, M. Brehm, R. Schade, O. Schütt, J. V. Pototschnig, H. Mirhosseini, A. Knüpfer, D. Marx, M. Krack, J. Hutter, and T. D. Kühne, The cp2k program package made simple, *The Journal of Physical Chemistry B* **130**, 1237 (2026).
- [7] S. P. Huber, E. Bosoni, M. Bercx, J. Bröder, A. Degomme, V. Dikan, K. Eimre, E. Flage-Larsen, A. Garcia, L. Genovese, D. Gresch, C. Johnston, G. Petretto, S. Poncé, G.-M. Rignanese, C. J. Sewell, B. Smit, V. Tseplyaev, M. Uhrin, D. Wortmann, A. V. Yakutovich, A. Zadoks, P. Zarabadi-Poor, B. Zhu, N. Marzari, and G. Pizzi, Common workflows for computing material properties using different quantum engines, *npj Computational Materials* **7**, 136 (2021).
- [8] E. Bosoni, L. Beal, M. Bercx, P. Blaha, S. Blügel, J. Bröder, M. Callsen, S. Cottenier, A. Degomme, V. Dikan, K. Eimre, E. Flage-Larsen, M. Fornari, A. Garcia, L. Genovese, M. Giantomassi, S. P. Huber, H. Janssen, G. Kastlunger, M. Krack, G. Kresse, T. D. Kühne, K. Lejaeghere, G. K. H. Madsen, M. Marsman, N. Marzari, G. Michalícek, H. Mirhosseini, T. M. A. Müller, G. Petretto, C. J. Pickard, S. Poncé, G.-M. Rignanese, O. Rubel, T. Ruh, M. Sluydts, D. E. P. Vanpoucke, S. Vijay, M. Wolloch, D. Wortmann, A. V. Yakutovich, J. Yu, A. Zadoks, B. Zhu, and G. Pizzi, How to verify the precision of density-functional-theory implementations via reproducible and universal workflows, *Nature Reviews Physics* **6**, 45 (2024).
- [9] L. Zhang, A. Kozhevnikov, T. Schulthess, H.-P. Cheng, and S. B. Trickey, Performance en-

- hancement of apw+lo calculations by simplest separation of concerns, *Computation* **10**, 43 (2022).
- [10] L. Zhang, A. Kozhevnikov, T. Schulthess, S. B. Trickey, and H.-P. Cheng, All-electron apw+lo calculation of magnetic molecules with the sirius domain-specific package, *The Journal of Chemical Physics* **158**, 234801 (2023).
 - [11] SIRIUS developers, SIRIUS: Domain specific library for electronic structure calculations (2026), documentation and source repository, <https://electronic-structure.github.io/SIRIUS-doc/> and <https://github.com/electronic-structure/SIRIUS>.
 - [12] K. G. Dyall and E. van Lenthe, Relativistic regular approximations revisited: An infinite-order relativistic approximation, *The Journal of Chemical Physics* **111**, 1366 (1999).
 - [13] J. Hutter and T. M. A. Müller, CP2K calculations for structures of the small molecule database with the revised MOLOPT basis sets and pseudopotentials, Zenodo (2023), version 1.0, dataset.
 - [14] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Physical Review Letters* **77**, 3865 (1996).
 - [15] C. Adamo and V. Barone, Toward reliable density functional methods without adjustable parameters: The pbe0 model, *The Journal of Chemical Physics* **110**, 6158 (1999).
 - [16] J. Tao, J. P. Perdew, V. N. Staroverov, and G. E. Scuseria, Climbing the density functional ladder: Nonempirical meta-generalized gradient approximation designed for molecules and solids, *Physical Review Letters* **91**, 146401 (2003).